

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Files and folders are organized logically.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables are named based on their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are descriptive and easy to understand.
5	Code Comments	Inline and block comments explain logic	Partial	Comments are added for important sections and complex logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code is divided into reusable functions/modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unused code has been removed.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic exception handling is implemented for user inputs and prediction errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (Pandas)	Data Cleaning & Feature Engineering	High

2	Machine Learning Model	Scikit-learn	Credit Card Approval Prediction	High
3	Prediction Function	Python	Predicting Approval Status	High
4	User Interface	Streamlit	User Input & Prediction Display	Medium
5	Data Visualization	Matplotlib	Result Analysis	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with proper modules.
Readability	5	Easy to understand with meaningful naming conventions.
Reusability	5	Functions and modules are reusable across the project.
Documentation / Comments	4	Adequate comments provided for key sections.
Overall Score	4.8/5	Clean, modular, and maintainable code following good coding practices.